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SECP3213-02 - BUSINESS INTELLIGENCE

PHASE 1 : PROJECT PROPOSAL

TITLE : Retail Sales Performance and Customer Behaviour Analysis

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1. Problem Background

1.1 Business contact

In this era, online shopping is growing faster like the internet, creating a massive amount of data on everything from customer orders to shipping and reviews. However, a lot of companies still struggle in using the information to make a smart business decision. To overcome this problem, this project explores the Brazilian e-commerce market using the Olist dataset. Olist is a real platform that helps small businesses sell their products on larger websites. Their dataset is incredibly useful because it includes over 100,000 actual orders placed between 2016 to 2018, covering customer details, product types, payment methods, reviews, and delivery times. This dataset perfectly shows a real company's system and it is a good starting point for building a Business Intelligence Project.

1.2 Importance of the problem

Despite that a lot of data is been generate, online stores still struggle to make good decisions in a few key areas:

- Focusing On Volume Over Profit

It is hard to see which products are actually making money. This lead companies to over-invest in items that sell a lot but do not actually bring in much profit

- Losing Existing Customer

It is harder to keep the customer that we already have rather than finding a new customer. Without tracking this properly, marketing money will be wasted to something that are not effective

- Missing Hidden Problem

Warning signs such as bad reviews or low ratings often get buried in the raw numbers. Because of this, the manager sometime miss important issue about poor product quality or shipping delays where it can lead to customer lost

A good Business Intelligence system solves this problem by optimizing the data that they have into a clear, easy to understand picture.

1.3 Expected impact

Here is how the proposed Business Intelligence system will directly help the business :

- Focusing on what actually make money :
By showing a true profit for each type of product, The manager can easily focus on the most profitable products that are underperforming
- Keeping customers coming back :
By looking at the customer behavior or habits such as what product that people often or recently buy, the system can spot customers who might be losing interest. This can help the business to make an offer to win them back before they leave.
- Fixing Hidden Problem :
By linking the bad review and low ratings product, the company can take quick action on handling the poor service and fix the root cause.

2. Objective

This project aims to achieve the following two objectives :

1. Boosting Sales and Profits :
Figure out exactly which products and locations are bringing in the most revenue and actual profit. This will help the business decide what items to stock, where to focus their marketing, and how to set better prices.
2. Understanding and Keeping Customer :
Group shoppers based on their buying habits such as how recently, how often and how much they buy. This sorts them into clear categories so the business can figure out exactly why people are leaving and how to keep them coming back.
3. Learning from Customer Feedback :
Dig Deeper into the customer review score to see why people leave a bad review. This will help businesses pinpoint the exact cause to help the company fix the most frustrating parts of the shopping experience.

3. Datasets Description

Dataset 1 - Orders Dataset

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Dataset
Files	olist_orders_dataset.csv
Data type	Structured (CSV)
Records	~99,000 orders
Key features	order_id, customer_id, order_status, order_purchase_timestamp, order_delivered_customer_date

Dataset 2 - Order Items Dataset

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Dataset
Files	olist_order_items_dataset.csv
Data type	Structured (CSV)
Records	~113,000 order items
Key features	order_id, order_item_id, product_id, seller_id, shipping_limit_date, price, freight_value

Dataset 3 - Customer Reviews Dataset

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Dataset
Files	olist_order_reviews_dataset.csv
Data type	Structured (CSV)
Records	~100,000 reviews
Key features	review_id, order_id, review_score, review_comment_title, review_comment_message, review_creation_date, review_answer_timestamp

Dataset 4 - Products Dataset

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Dataset
Files	olist_order_products_dataset.csv
Data type	Structured (CSV)
Records	~32,000 products
Key features	review_id, order_id, review_score, review_comment_title, review_comment_message, review_creation_date, review_answer_timestamp

Dataset 5 - Seller Dataset

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Dataset
Files	olist_sellers_dataset.csv
Data type	Structured (CSV)
Records	~3,000 sellers
Key features	seller_id, seller_zip_code_prefix, seller_city, seller_state

Dataset 6 - Payment Dataset

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Dataset
Files	olist_order_payments_dataset.csv
Data type	Structured (CSV)
Records	~104,000 sellers
Key features	order_id, payment_sequential, payment_type, payment_installments, payment_value

Dataset 7 - Category Translation Dataset

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Dataset
Files	olist_order_payments_dataset.csv
Data type	Structured (CSV)
Records	~71 translations
Key features	product_category_name, product_category_name_english

4. Proposed BI Approach

4.1 Data Integration Methods

4.1.1 Schema Reconciliation

Each CSV file in sources has a distinct schema which has different column sets, data types and levels of granularity. The Select tool in Alteryx is used to rename ambiguous columns like price in the order items table renamed to item_price to avoid collision with any seller-level price fields and to verify that join keys are of matching data types before any join is executed.

4.1.2 Granularity Alignment

The orders table is at order level (one row per order) while the order items table is at line-item level (a single order with three items produces three rows). Before joining these to the reviews table which is also at order level, the Summarize tool is used to aggregate the order items table up to order level, computing total revenue (sum of item_price), total freight (sum of freight_value) and item count (count of order_item_id) per order. This solves the granularity mismatch problem and ensures a one-to-one join with the orders table.

4.2 Data Cleaning Methods

4.2.1 Null Value Treatment

Null handling strategy is applied using Alteryx's Data Cleansing and Formula tools based on this type of null:

- Numeric nulls: Replaced with the column median computed using the Summarize tool.
- Categorical nulls: Replaced with the label 'Unclassified' to preserve record count without affecting category aggregations.
- Timestamp nulls: Excluded from delivery duration calculations via a Filter tool condition.

4.2.2 Data Type Standardisation

All timestamp fields such as `order_purchase_timestamp`, `order_delivered_customer_date` and `review_creation_date` are converted from string format to DateTime type using Alteryx's DateTime tool. This enables Power BI to correctly place these fields on a time axis and supports the usage of duration fields.

4.2.3 Outlier and Anomaly Filtering

Orders with a status of 'cancelled' or 'unavailable' are removed using the Filter tool to ensure that revenue and performance metrics reflect only completed commercial transactions. Records with a price value of zero also are excluded from all financial analysis.

4.3 Data Transformation Methods

4.3.1 Derived Calculated Fields

The following calculated fields are derived using Alteryx's Formula tool after the join is complete:

- Delivery duration (days): `order_delivered_customer_date` minus `order_purchase_timestamp` then converted to days.
- Total order value: `item_price` plus `freight_value` per order.

- Review-to-delivery lag (days): $\text{review_creation_date} - \text{order_delivered_customer_date}$.
- Profit proxy: $\text{item_price} - \text{freight_value}$.

4.3.2 RFM Score Derivation

Customer RFM scores are derived through a three-step process in Alteryx:

1. Summarize tool groups all orders by customer_id, computing into three new columns which are days since last order (Recency), total number of orders (Frequency) and total spend (Monetary).
2. The formula tool assigns a score of 1 to 5 to each dimension using quintile thresholds.
3. A composite RFM segment label is derived like '555' = Champion and '111' = Lost Customer using a nested IF formula. This produces a categorical segmentation field for use in Power BI.

4.4 Data Visualization

Five visualisation forms are planned in Power BI for different analytical objective which are:

Visual form	Analysis
Bar chart	Sales revenue and profit by product category and seller state, ranked to identify top and bottom performers.
Line Chart	Monthly order volume and total revenue over time, with trend line overlay to identify seasonality and growth trajectory.
Heatmap	RFM customer segmentation matrix plotting Recency against Frequency, with Monetary value encoded as colour intensity. Segments customers into actionable groups.
Scatter plot	Relationship between freight value and review score, to identify whether higher delivery costs correlate with lower customer satisfaction.
Treemap	Distribution of low review scores (1–2 stars) across product categories, sized by order volume. Identifies product lines with disproportionate dissatisfaction relative to sales volume.